Customer Sales Analysis

Customer shopping behaviour, purchasing patterns and category performance

Project type	Portfolio project analysing a public customer-shopping dataset; not real client data
Tools	SQL (MySQL), Python (Pandas/Jupyter Notebook), Microsoft Power BI
Data	3,900 customer records; purchase amounts in USD; customer, product, category, subscription, shipping and review attributes
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1. Executive summary

This project explores customer shopping behaviour using a three-part workflow: Python prepares and engineers the data, SQL answers targeted business questions, and Power BI presents the results in an interactive dashboard. The notebook contains 3,900 records, with a mean purchase amount of **$59.76** and an average review rating of **3.75/5** after missing review ratings are filled using category-level medians. The supplied dashboard highlights category revenue, category sales, subscription status and revenue/sales across age groups.

Headline figures shown in the supplied dashboard

3.9K	$59.76	3.75	27%
Customer records	Average purchase amount	Average review rating	Subscribers

Interpretation note: Dashboard labels and rounded chart values are transcribed from the screenshot. The raw CSV and Power BI source file were not included with the uploaded materials, so detailed chart values could not be independently recalculated here.

2. Business problem

Retail teams need to understand who is buying, which categories generate the most revenue, how subscription status relates to customer behaviour, and whether discounts, shipping choices and repeat purchasing patterns reveal useful segments. The original data is row-level, so the analysis converts it into summary measures and a dashboard for faster review.

Business questions explored

• How much purchase revenue is represented in the data, and what is the average purchase amount?

• How do category revenue and sales counts compare?

• What share of customers are subscribers versus non-subscribers?

• How do revenue and sales differ across age groups?

• Do purchase amount, gender, discount use, shipping type and previous purchases reveal useful patterns?

• Which products have strong review ratings or high discount usage?

3. Dashboard screenshot

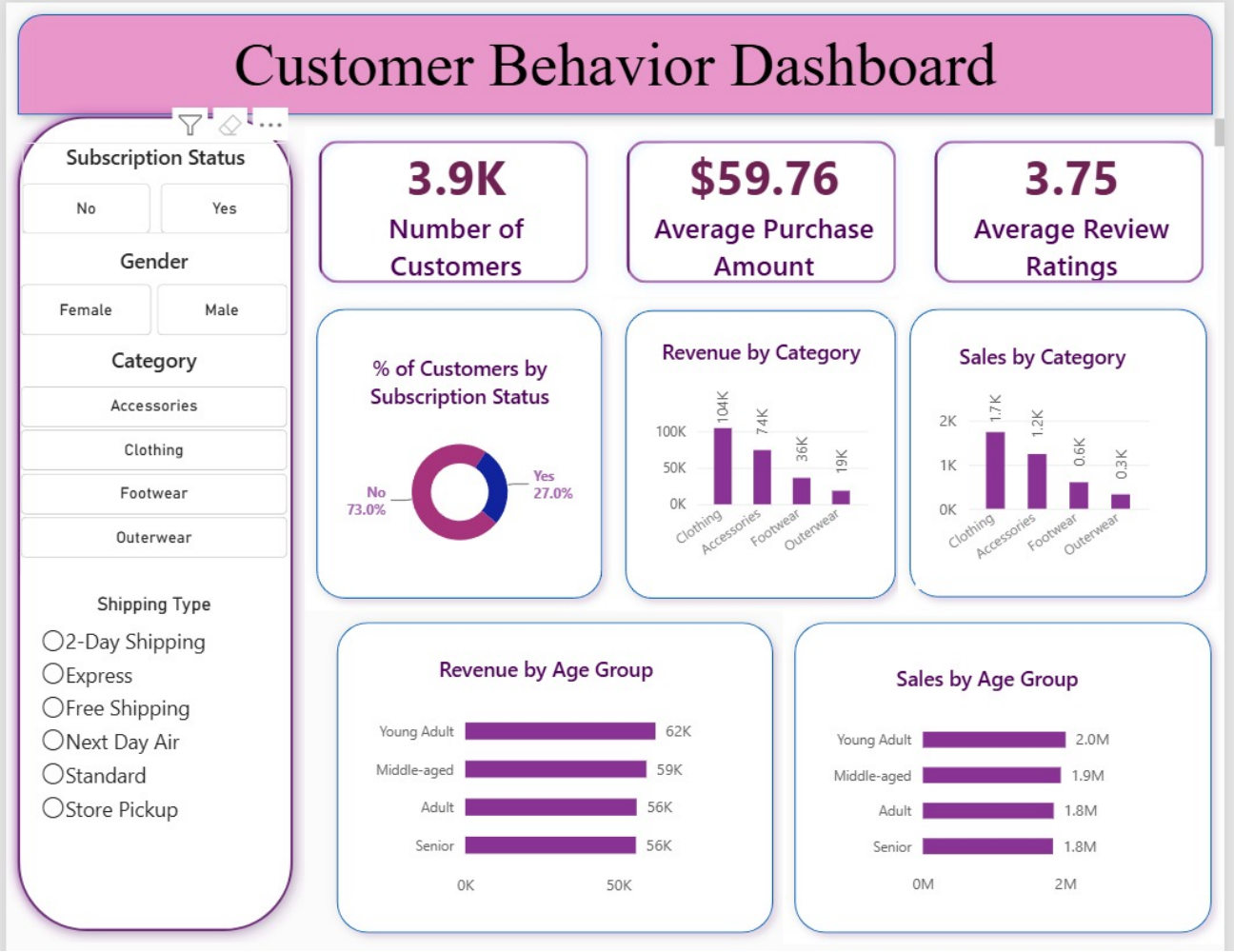


Figure 1. Customer Behavior Dashboard supplied for this project. The dashboard includes slicers for subscription status, gender, category and shipping type; KPI cards; and charts for subscription share, category revenue/sales, and age-group comparisons.

4. Dataset and preparation

Item	Details from the supplied notebook
Rows	3,900 records; customer IDs run from 1 to 3,900 in the displayed sample and summary output.
Fields	Age, gender, item purchased, category, purchase amount, location, size, color, season, review rating, subscription status, shipping type, discount applied, previous purchases, payment method and purchase frequency.
Missing data	The notebook reports 37 missing Review Rating values and no missing values in the other displayed columns. Ratings are filled with the median rating within each category; the notebook then reports zero missing ratings.
Feature engineering	Age is binned into four quantile-based groups: Young Adult, Adult, Middle-aged and Senior. Purchase frequency labels are mapped to approximate day counts.
Redundant field	The notebook checks that Discount Applied and Promo Code Used match in all rows, then drops Promo Code Used.
SQL load	The notebook writes the processed DataFrame to a MySQL table named customer in the customer_shopping_da4 database.

5. Solution process

1. Inspect the data — Load the CSV with Pandas; review columns, data types, descriptive statistics and missing values.

2. Clean and standardise — Impute missing review ratings by category median, standardise column names and rename purchase_amount_(usd) to purchase_amount.

3. Engineer features — Create age_group and purchase_frequency_days to support segment-level comparisons.

4. Load to MySQL — Use SQLAlchemy with the MySQL connector to replace/write the processed table and verify a small sample read-back.

5. Query business questions — Use aggregation, filtering, common table expressions and a window function to examine gender, discounts, ratings, shipping, subscription, customer segments, product rankings and age groups.

6. Visualise in Power BI — Use the supplied dashboard screenshot as the presentation layer for KPIs, slicers and category/age-group charts.

Reproducibility note: the uploaded notebook contains a local MySQL connection example and a hard-coded password. Do not publish real credentials in a public repository; replace them with environment variables or a local .env file excluded from version control.

6. Key findings

1. Purchase amount is tightly bounded in the sample.

The notebook summary reports purchase amounts from $20 to $100, with a mean of $59.76 and median of $60. This is a descriptive summary of the supplied dataset, not a forecast.

2. Review ratings need explicit treatment.

There are 37 missing review ratings in the raw notebook output. Category-median imputation removes the nulls, but it can compress variation and should be documented whenever rating averages are reported.

3. Subscription status is a visible dashboard segment.

The screenshot shows 27.0% Yes and 73.0% No. This is a useful baseline for segmenting customer behaviour, but it does not by itself establish that subscriptions cause higher spending.

4. Clothing leads category revenue in the dashboard.

The displayed chart labels are approximately Clothing 104K, Accessories 74K, Footwear 36K and Outerwear 19K. Values are rounded visual labels and should be rechecked against the Power BI model or SQL output before being quoted as exact figures.

5. Age-group revenue is relatively close in the screenshot.

The chart shows about 62K for Young Adult, 59K for Middle-aged, and 56K each for Adult and Senior. This suggests no single age group dominates the displayed revenue measure.

6. The SQL file is designed around actionable questions.

Queries cover discount use among above-average spenders, average ratings by product, Standard versus Express shipping, customer segmentation by previous purchases, top products within category, repeat-buyer subscription status and revenue by age group.

Selected figures visible in the dashboard

Dashboard measure	Displayed value	Source / qualification
Customer records	3.9K	KPI card; notebook has 3,900 rows
Average purchase amount	$59.76	Matches notebook descriptive mean ($59.764...)
Average review rating	3.75	Notebook mean before imputation shown as 3.750; dashboard appears rounded to 2 decimals
Subscription status	Yes 27%; No 73%	Percent labels visible in donut chart
Category revenue	Clothing 104K; Accessories 74K; Footwear 36K; Outerwear 19K	Approximate chart labels; units/aggregation should be confirmed in the PBIX model

7. Recommendations

1. Prioritise category-level planning. Clothing and Accessories lead the displayed category revenue chart. Check margins and inventory costs before allocating more stock or marketing spend.

2. Investigate subscription conversion. Compare average spend, purchase frequency and repeat-purchase behaviour for subscribers and non-subscribers. Use customer-level data and avoid treating a cross-sectional association as causation.

3. Use discounts with a clear measurement plan. Track discount rate by product and compare purchase amount and repeat behaviour for discounted versus non-discounted records. Evaluate margin impact if cost data becomes available.

4. Segment retention efforts by purchasing history. The SQL query groups customers into New, Returning and Loyal segments based on previous purchases. Confirm the thresholds and define what 'new' means before operational use.

5. Validate all dashboard measures against SQL. Export the Power BI aggregations or run matching SQL queries so category, age-group, and subscription figures reconcile before publishing final portfolio claims.

8. Data notes and limitations

• The report is based on the uploaded notebook, SQL script and dashboard screenshot. The raw CSV and editable Power BI (.pbix) file were not supplied, so most chart aggregates could not be independently recalculated.

• The notebook fills missing ratings with category medians. This is a modelling/preparation decision and may influence downstream averages.

• The dataset appears to contain one row per customer record with purchase-related attributes; do not assume it represents a complete transaction history unless the source data dictionary confirms that interpretation.

• The SQL query labelled as comparing subscription revenue currently selects AVG(purchase_amount) for both avg_spend and total_revenue. The total_revenue expression should be SUM(purchase_amount) before using that query to make a total-revenue claim.

• The SQL script's New segment is defined as previous_purchases = 1. If zero previous purchases is possible, it would not be counted as New under the current logic.

• The notebook includes a hard-coded database password. Remove it from the public version and rotate it if it is a real credential.

• No product cost, margin, acquisition cost or longitudinal outcome data is shown; recommendations should be treated as hypotheses to validate, not causal conclusions.

9. Skills demonstrated

Data cleaning and missing-value handling | Pandas descriptive analysis | Feature engineering | MySQL table loading | SQL aggregation and filtering | CTEs and window functions | Customer segmentation | Power BI dashboard interpretation | Analytical communication and data-quality review

Note: This is an independent portfolio project based on a customer-shopping practice dataset. Figures explicitly identified as dashboard labels are approximate unless independently confirmed by the notebook or SQL output.